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Chapter 32: The Great Acceleration—AI's Material Substrate, Positive Feedback Loops, and Planetary Boundaries

Abstract

This chapter examines the energy, water, and emissions consequences of AI infrastructure expansion, distinguishing the evidentiary strength of corporate disclosures, scenario estimates, and mechanism-level inferences, and noting that the scarcity of data is both a component of the externality structure and an evidentiary limitation of this chapter. On this evidentiary basis, it identifies four reinforcing loops (capability–demand, AI-accelerated R&D, economic reinvestment, and infrastructure lock-in), one efficiency–rebound mechanism (the Jevons mechanism), and three balancing loops (price, regulation, and planetary boundaries), and argues that their time scales are asymmetric.

The scale of analysis must therefore expand from the single query to the infrastructure layer: when the cadence of reinforcing loops outpaces the response of balancing loops and multi-year lock-in deposits their amplification into stock, overshoot becomes a risk scenario to be tested rather than a foregone conclusion. The tobacco analogy fractures at the applicability of governance tools—linear extrapolation from per-use costs alone is insufficient to infer AI's system-level consequences—synchronized loads, investment inertia, and demand feedback require the scale of governance to extend to the infrastructure and investment levels, and the software-engineering literature together with historical cases independently attests the mechanism that continued output does not establish output reliability. Carbon pricing exposes, across attribution, time, and boundaries, the limits of any single

accounting metric for establishing all environmental consequences or justifying deployment; path dependence and tipping risk support constraints beyond pricing without establishing that marginal costs tend to infinity.

The chapter uses "burning" for these material costs obscured by the interface—the oxidation of fossil carbon, the fission of uranium, and the manufacture of silicon, three physical processes on one and the same externalized cost chain—and closes on the question that truly needs inventorying: which combustion purchases genuine output, and which merely accelerates the burning of the Earth to produce hallucinations, sycophancy, and other pseudo-productivity.

Keywords

AI infrastructure; environmental footprint; feedback loops; rebound effect; infrastructure lock-in; planetary boundaries; environmental accounting; mapping element (token)

§1 The Invisible Burning

This sentence is burning.

This is not merely a metaphor. In a physical sense: the text you are reading, at the moment it was generated, consumed electricity and dissipated heat—and, following the cooling configuration of the data center, consumed or evaporated water, and, following the generation mix of the supplying grid, emitted carbon into the atmosphere. These physical events occurred in invisible places—on some row of servers in some data center, in some transmission line of some power grid, on the surface of some river or reservoir. They will not appear on the screen, nor in the bill (the subscription fee paid for AI has no line item called "carbon emissions"), nor in the AI's own replies.

But there is a crucial difference: human respiration is part of the biosphere's carbon cycle; the exhaled CO_2 will be re-fixed by plants. Where fossil energy supplies the computation, AI operation adds long-stored carbon to the active carbon cycle: natural sinks absorb part of it only slowly, and its climatic effects persist on a centennial scale. Every inference has a material cost, but its specific carbon and water consequences depend on the power supply, cooling method, and accounting boundary.

Chapters 28 and 31 of this book addressed, respectively, the cognitive and cultural layers of this system; this chapter turns to its obscured material layer. Chapter 10 of this book already used tobacco as a reference case for AI risk—harmful products, concealed hazards, lagging regulation—and addressed the continuity of the addiction dimension within the limbic capitalism lineage of Courtwright (2019). This chapter extends the analogy to the full dimensions of externalities: concealing known risks, shifting costs onto the public health system, locking in users through addiction mechanisms, and using the discourse of "personal choice" to resist regulation. However, the core argument of this chapter is: the tobacco analogy fails on one crucial dimension, and precisely that point of failure is the heart of the matter.

Tobacco's externalities are additive. When one hundred million people smoke, the externality is roughly one hundred million times that of one person smoking. The total harm can be reduced linearly by reducing the number of smokers. The leverage of regulation operates at the individual node: raise taxes, ban advertising, label warnings—each measure reduces the total by altering individual behavior. This is a linear intervention in a linear system, difficult but conceptually clear.

AI's externalities are not additive. They are coupled. The physical cost of AI use by one hundred million people cannot be inferred by simply multiplying a single user's cost by one hundred million. Infrastructure scale effects, feedback loops, and the recursive structure of AI accelerating its own development can make that relationship nonlinear; its functional form depends on the deployment structure and investment dynamics. In this system, reducing individual usage does not linearly reduce total harm, because the primary driver of harm is not the sum of individual consumption but the dynamics of infrastructure expansion. The leverage of regulation is drastically weakened at the individual node—one can make a person use AI less, but one hundred million people each using it a little less may not change committed training and construction decisions in time, because those decisions occur on an entirely different level. The tobacco comparison in this chapter uses a simplified consumption-side model of tobacco governance and does not claim that the tobacco industry lacks systemic environmental externalities.

The point at which the tobacco analogy breaks down—the transition from linear to nonlinear externalities—is not a precise threshold but a structural phase change. When AI's physical footprint transforms from "the pollution problem of one industry" into "a systemic driver of planetary boundaries," the scale of analysis and the mix of tools need to expand. This chapter attempts to provide the basic components of that tool set.

§2 Three Indicators and Their Shadows

To understand AI's physical cost, at least three indicators must be tracked: electricity consumption, water resource consumption, and carbon emissions. The data situation of these three indicators is itself part of the diagnosis.

Electricity. The International Energy Agency (IEA) estimated in its *Electricity 2024* report that data centers (including AI and cryptocurrency) consumed a combined total of approximately 460 terawatt-hours (TWh) in 2022 and could reach 620–1,050 TWh by 2026 (IEA, 2024). A thematic report, *Energy and AI*, released by the IEA in April 2025, gives a data-center-only estimate: global data center electricity consumption in 2024 was approximately **415 TWh**, accounting for approximately **1.5%** of global electricity consumption—the two figures differ in statistical boundary and cannot be joined into a single time series; under its baseline scenario, data center electricity consumption will grow at an average annual rate of approximately **15%** from 2024 to 2030 (IEA, 2025). AI is rapidly becoming the dominant factor in this growth—Goldman Sachs (2025) predicts that global data center power demand will increase by **50%** by 2027 compared to 2023 levels and by **165%** by 2030 (Goldman Sachs, 2025).

Within the United States itself, the data show this change most directly: according to the revised historical series published by Lawrence Berkeley National Laboratory (LBNL) in 2024 under commission from the U.S. Department of Energy (DOE), data center electricity consumption rose from approximately **60 TWh** in 2014 to **176 TWh** in 2023, nearly tripling, and is projected to reach **325–580 TWh** by 2028, accounting for **6.7–12%** of total U.S. electricity consumption at that time (Shehabi et al., 2024, *2024 United States Data Center Energy Usage Report*).

Running parallel to this is a quantum leap in the scale of individual facilities: the individual scale of traditional cloud data centers is typically below 100 MW, whereas the 2025 large-load white paper of the North American Electric Reliability Corporation (NERC) discusses Meta's planned Louisiana Richland Parish project at approximately **2 GW** (NERC, 2025a), while OpenAI Stargate's 10 GW is a multi-site aggregate commitment

that cannot be taken as a single campus's capacity (OpenAI project disclosures, as cited in Bashir et al., 2026). Bashir et al. estimate the relative scale of training campuses against California load—their citation is California Independent System Operator (CAISO) 2025 real-time data, and the paper synthesizes Meta's single-campus disclosure and the OpenAI Stargate multi-site program into a range of "2–10 GW, **6–32%** of California's load per site," a synthesized range whose statistical objects are inconsistent (Bashir et al., 2026, §1, citing CAISO 2025). For comparison, 1 GW is approximately **1.7 times** the average electricity demand of the city of San Francisco (Chung et al., 2026).

This scale leap has already produced observable grid-side consequences: in **July 2024**, a transmission fault in Northern Virginia caused approximately **1.5 GW** of voltage-sensitive data center load to collectively disconnect from the grid within seconds—the circuit breakers at the data center substations did not trip; the load loss resulted from customer-side protection and control actions, with some facilities switching to backup power; NERC compared this event to "the abrupt loss of a large nuclear plant" (Bashir et al., 2026, §1, relaying NERC's assessment; the event is documented in NERC, 2025a). NERC subsequently issued two formal alerts in the domain of large load unexpected disconnection and computational load reliability: a Level 2 Alert on September 9, 2025 (addressing multiple "widespread and unexpected" disconnection events of large loads above 1,000 MW in 2024–2025) and a Level 3 Essential Actions Alert on May 4, 2026 (covering computational loads including AI data centers, the highest-level alert issued by NERC in this risk domain) (NERC, 2025b; NERC, 2026). David Penny, Director of Reliability Services for the Texas Reliability Entity (Texas RE), in public discussion directly analogized the AI training load profile to a steel mill—"very fast, very large ramps" (Walton, 2025, *Utility Dive*; also cited by Bashir et al., 2026, §1 and §3).

In the power engineering literature, AI training clusters, due to tens of thousands of GPUs starting the same training job in lockstep, exhibit the characteristics of a "single correlated load"; the closest historical precedents for their scale and temporal behavior are heavy industrial facilities such as aluminum smelters and semiconductor fabrication plants—the latter were absorbed by the grid through customized interconnection protocols, wholesale electricity price exposure, and emergency load-shedding clauses, but planned capacities are entering the gigawatt range, while the rapid power swings of synchronized training workloads and the common-mode disconnection of large data-center loads have each become reliability risks requiring separate study—the former a workload behavior, the latter a protection-and-control response to grid disturbances; the two should not be treated as the same kind of event (Bashir et al., 2026, §3). These figures and events collectively indicate: AI data centers are no longer a minor load on the grid, but a new type of load that has breached the century-old assumptions of grid planning in terms of scale, temporal characteristics, and spatial concentration.

However, "total data center energy consumption" is a crude proxy. The indicator that is truly needed but not yet reliably available is: the electricity consumed respectively by AI training and inference and their growth trajectories. Existing scattered estimates include: the training of GPT-3 was estimated to consume approximately 1,287 megawatt-hours (MWh) of electricity and emit approximately 552 tonnes of CO_2 equivalent (Patterson et al., 2021). The energy consumption of training GPT-4 is widely believed to be an order of magnitude higher; OpenAI has not publicly disclosed the precise figure. The energy consumption on the inference side is even harder to track, and its authoritative estimate itself underwent a significant downward revision—De Vries (2023, *Joule*) initially estimated, based on early 2023 hardware and parameter assumptions, that a single ChatGPT query consumes approximately 3 Wh of electricity (roughly 10 times that of a single Google search); this figure was widely cited. Epoch AI (2025), after re-estimating using GPT-4o, NVIDIA H100, and a token count more

representative of actual loads, revised the electricity consumption of a typical ChatGPT query downward to approximately **0.3 Wh/query**, i.e., about one-tenth of the old estimate—though the two estimates rest on different models, hardware, and workload assumptions (Epoch AI, 2025).

The key methodological lesson here is: inference energy consumption evolves rapidly with models and hardware; any narrative claiming a stable ratio of "N times a Google search" must be annotated with the model generation and measurement year on which it is based—this is also a concrete example of the "impossible accounting" discussed later in this chapter (§7). More crucially, as AI is embedded into an increasing number of everyday applications—search, email, document editing, programming, image generation—the cumulative energy of inference can exceed the one-time cost of a single training run within months once deployment scales (Stanford HAI, 2026); the relative shares of training and inference, however, depend on deployment scenarios—LBNL's model even projects training server energy to surpass inference by 2028 (Shehabi et al., 2024). This chapter does not rest its argument on which of the two grows faster. When the total volume of inference itself is driven by positive feedback loops, total inference energy depends jointly on usage volume, task composition, and energy per task; per-query efficiency gains alone do not guarantee a decline in aggregate consumption—and this is precisely where the tobacco analogy begins to fail.

Water. Data centers require large volumes of cooling water. According to Microsoft's 2024 Environmental Sustainability Report, Microsoft's global water consumption in FY23 was approximately **7.84 million cubic meters**, an increase of approximately 87% over the FY20 baseline (approximately 4.2 million cubic meters) (Microsoft, 2024). Google's 2024 Environmental Report disclosed that its global data centers consumed approximately **6.1 billion U.S. gallons** (approximately 23.1 billion liters) of water in 2023, an increase of 17% over the previous year, an increase the report attributes to the workload and cooling demands brought by the expansion of AI products and services (Google, 2024). Special attention must be paid to units and scopes: the two companies' disclosures are in cubic meters and gallons, respectively; when comparing directly, conversion and verification of the distinction between "withdrawal / consumption / evaporation" are required. Such ambiguity in units and scopes is itself part of the data opacity diagnosed in §2. The study by Li et al. (2025) estimated that, under the U.S. data center conditions it assumed, training GPT-3 involved approximately **700,000 liters** of on-site water consumption, and approximately 5.4 million liters including water consumption associated with electricity generation; a conversation with GPT-3 involving 10–50 questions consumes approximately **500 milliliters** of water—roughly equivalent to a bottle of mineral water, a figure that varies with deployment location and time (Li et al., 2025).

The problem with water consumption lies not only in the total volume but also in the geographical injustice of its distribution: data centers are often built in areas with cheap electricity, which may simultaneously be under water stress. AI's water footprint is a severely underestimated externality, partly because there is a systematic geographical and socioeconomic distance between the bearers of water crises (farmers in arid regions, communities dependent on groundwater) and the beneficiaries of AI (global middle-class consumers using AI products). The harm occurs at one end of the supply chain; the pleasure occurs at the other; and there is no feedback loop between the two ends—the voices of the victims are not in the AI's training data, nor in the product managers' OKRs.

Carbon. The carbon footprint of training GPT-3 was estimated at approximately **552 tonnes of CO_2 equivalent** (Patterson et al., 2021, using the average U.S. grid carbon intensity). But the significance of this figure depends on the grid's carbon intensity, which varies by region and time period—the carbon footprint of the same data

center can differ by a factor of several times when running on solar power during the day versus on a natural gas peaker plant at night. A more macro-level trend: despite major technology companies having made "net-zero" pledges, their actual carbon emissions have continued to rise during the AI expansion. Google disclosed in its 2024 Environmental Report that its company-wide greenhouse gas emissions in 2023 had risen by **48%** compared to the 2019 baseline, an increase driven primarily by data center energy consumption and supply chain emissions; the report further notes that as AI is more deeply integrated into products, the energy demand driven by greater AI compute intensity will make emission reductions more difficult (Google, 2024). Microsoft itself disclosed in its 2024 Environmental Sustainability Report that its full-scope Scope 1–3 emissions had risen by approximately 29.1% since the 2020 baseline (Microsoft, 2024). The divergence between "net-zero" pledges and actual emission curves is not merely corporate-level saying one thing and doing another (though it is also that), but a symptom of a structural fact: the expansion speed of AI infrastructure exceeds the speed of renewable energy deployment and carbon offset mechanisms. One can simultaneously sincerely commit to net-zero and genuinely increase emissions—as long as business growth outpaces emission reductions, and this is precisely what the AI industry is doing.

The Stanford HAI 2026 AI Index provides the most recent estimates up to 2025: xAI's Grok 4 has an estimated training emission of **72,816 tonnes of CO_2 equivalent**—per the Stanford AI Index's original comparison, **over a thousand times the lifetime carbon emissions of an average car (approximately 63 tonnes of CO_2 equivalent)**. For comparison, GPT-4 is estimated at **5,184 tonnes**; Meta's Llama 3.1 405B at **8,930 tonnes** (Stanford HAI, 2026: §1.4, Figure 1.4.3). The roughly 14-fold jump from GPT-4 to Grok 4 occurred within roughly two years. AI data center power capacity rose to approximately **29.6 GW** by Q4 2025—per the Stanford AI Index's original comparison, roughly equivalent to the peak electricity demand of New York State (Stanford HAI, 2026: §1.2, Figure 1.2.4). The annual water consumption of GPT-4o inference alone is estimated in the range of **1.3 to 1.6 million kiloliters**, exceeding, at the high-end estimate, the annual drinking water needs of **1.2 million people** (Stanford HAI, 2026: §1.4, Figure 1.4.8).

These three indicators collectively constitute the basic outline of AI's material footprint. But a deeper methodological question lies: why are these data so difficult to obtain?

Part of the reason is technical—AI workloads and non-AI workloads are mixed within the same data center, making precise attribution difficult. But the more fundamental reason is structural: AI companies lack the incentive to disclose these data. Energy consumption and carbon emission data are still voluntarily disclosed in most jurisdictions; the EU has established mandatory reporting of energy and water data for large data centers since 2024 (European Commission, 2024); water consumption data often appears only in the appendices of corporate social responsibility reports; the most critical metric—the marginal physical cost per unit of AI inference—still lacks consistent public disclosure: some companies have published per-request measurements or allocated estimates for specific production environments—e.g., Elsworth et al.'s measurement of Gemini production traffic (approximately 0.24 Wh for the median text prompt)—but such mean or median metrics are not identical to marginal costs and do not yet provide consistent cross-model, cross-regional, and full-lifecycle comparisons (Elsworth et al., 2025). This opacity is structurally isomorphic with the tobacco industry's strategy concerning epidemiological data: not directly falsifying, but making the establishment of causal links as difficult as possible. Insufficient quantification significantly raises the difficulty of regulation and attribution, but does not make regulation impossible in principle. The scarcity of data is not merely an obstacle to research; it is itself part of the structure of the externality—and also an evidentiary limitation of this chapter.

§3 Four Loops and One Mechanism

The reason AI's physical footprint does not grow linearly is that it is driven not by a single causal chain but by a set of mutually coupled positive feedback loops. System dynamics provides the formal language for describing these loops. What follows identifies four main reinforcing loops (R), one efficiency–rebound mechanism (the Jevons mechanism, R2), and three balancing loops (B), and analyzes their relative strengths and time scales.

Reinforcing Loop R1: The Capability–Demand Spiral. Stronger AI models → more viable application scenarios → greater user adoption → greater computational demand → greater capital investment → stronger AI models. This is the most fundamental growth loop. Each generation of model capability improvement opens up applications previously infeasible (from text generation to multimodal reasoning to autonomous agents); each new application brings new user cohorts; each new user means more inference computation; and the growth in inference demand in turn drives investment in the next generation of even stronger models. The cycle of this loop is approximately 12 to 18 months (roughly corresponding to the iteration cycle of major models); the interplay of capability and demand may raise total computational investment, but generational inference-energy baselines cannot be determined directly from training-emission comparisons. (These cross-model training-emission estimates—the 5,184, 8,930, and 72,816 tonnes for GPT-4, Llama 3.1 405B, and Grok 4 listed in §2—show differences in scale between individual training runs; they do not establish a generational baseline for inference energy use, and the report itself notes that hardware efficiency, training duration, and the carbon intensity of the energy supply all affect emissions. Testing R1 requires linked evidence on user demand, query volume, energy per comparable task, and investment responses.)

Efficiency–Rebound Mechanism R2: The Jevons Mechanism. More efficient AI chips and algorithms → lower per-inference cost → more usage scenarios become economically viable → total computation volume increases → total energy consumption increases. This is a possible reappearance of Jevons Paradox in the AI domain: efficiency improvements may release suppressed demand by lowering marginal cost; when the released demand exceeds the savings per task, total consumption rises. The energy efficiency ratio (compute per watt) of each generation of NVIDIA GPUs is indeed improving, but the total deployment volume of GPUs is growing even faster—the growth of total computational demand may outpace improvements in energy efficiency per unit. This mechanism is particularly dangerous because it converts "good news" (efficiency improvements) into a bad outcome (increased total consumption), thereby making the narrative that "technological progress will solve the problem" itself part of the problem. Under the condition that rebound exceeds the energy saved, every paper announcing "AI inference efficiency improved by X times" is a leading signal that total energy consumption will rise.

The Jevons mechanism also has a dimension systematically neglected in current energy efficiency discussions: the computational work per query itself is not constant but is rising dramatically with the evolution of model architectures. The energy efficiency ratio (energy per token) from GPT-3 to GPT-4 may have improved, but from GPT-4 to o1/o3-class deep reasoning models, the total computation per query—including the hidden chain of thought that unfolds in the background, invisible to the user, potentially reaching tens of thousands of tokens—may have increased substantially (the magnitude depending on task type, inference budget, and hardware boundary). A portion of this hidden computation is not used for "thinking about the problem itself" but for satisfying alignment constraints: avoiding safety triggers, finding feasible solutions among multiple instructions, and conducting multiple rounds of self-review and correction of its own outputs. From an information-theoretic

perspective, this is additional computation expended to press down the uncertainty of the output space—in order to maintain the "orderliness" of the output (safe, helpful, harmless), the model consumes vast computational resources in the background; this chapter has not built a model to quantify the thermodynamic dissipation involved, and these resource consumptions are invisible in the final output.

This produces an ironic structure: alignment—the process of making AI "safe"—adds computational cost in safety implementations that require additional reasoning, review, or model calls; its marginal cost and reliability benefits should be assessed for each implementation. Chapter 10 of this book discussed the epistemological cost of algorithmic Xiangyuan; it is now evident that it also has a physical cost. Every additional layer of safety filtering, every round of background self-review, every "let me think again" hidden reasoning, silently consumes electricity, dissipates heat, and evaporates cooling water. Alignment is not free. Its price enters operating costs in the form of electricity bills, and enters the environment in the form of carbon and water—what is missing is not bookkeeping, but attribution to specific activities and their environmental consequences. The marginal size of these inputs and their relation to reliability benefits must be assessed for each implementation; their invisibility in the final output does not by itself make them pseudo-productivity. If the classic Jevons mechanism involves demand growth absorbing efficiency gains, a possible variant here involves increasing computational complexity absorbing those gains, with part of the driver of complexity growth coming from alignment itself.

Reinforcing Loop R3: AI Accelerating AI. AI-assisted AI research → faster model iteration → stronger AI capabilities → AI assisting AI research more deeply. This is the most structurally unique of all the loops, and the one that makes the externality structure of AI structurally different from that of all previous technologies. The steam engine could not design a better steam engine; the automobile could not drive into a factory to manufacture more automobiles; but AI is being systematically used to accelerate the development of AI itself—from code writing to hyperparameter search to architecture exploration to paper summarization (the contrast here concerns the human rate limit in the R&D loop, not the self-replication capability of machines). This means that AI is not merely a technology that consumes resources; it is also a technology that may accelerate the rate of its own resource consumption; if research acceleration is transmitted into deployment expansion, the rate of growth itself will also rise—the conditions for that transmission are given in §5.

Reinforcing Loop R4: The Economic Lock-in Loop. AI boosts productivity → GDP growth → more capital available for AI investment → larger-scale AI deployment → more productivity gains. AI's economic promise—boosting productivity across various industries—if realized, will generate more economic surplus, a substantial portion of which will be reinvested into AI infrastructure. The strength of this loop depends on AI's actual productivity effect (currently still under debate), but even if the effect is relatively modest, as long as it is positive and the reinvestment rate is high enough, it may sustain positive feedback. More importantly, this loop grants the expansion of AI a self-consistent economic legitimacy—"AI creates its own demand"—which makes any policy attempting to limit AI's scale appear, in economic discourse, as though it is hindering growth.

There are two layers of specific qualification that need to be marked. **The first layer is the heterogeneity of output nature.** The operation of the R4 loop does not presuppose output equivalence—revenue and reinvestment can continue even when output is overestimated; output equivalence is the premise for treating rising GDP as evidence of genuine productivity. If what AI actually contributes is a different kind of output—maintenance costs shifted off-system, outputs that appear to pass acceptance but are untraceable along causal chains—then the rise in GDP figures no longer corresponds to genuine productivity surplus, while R4's positive feedback in the physical energy consumption dimension still takes effect in full; §6 will independently address this distinction.

The second layer is the intertemporal mismatch of accounting. In the risk scenario where benefits are overstated and new deployment is used to postpone the exposure of existing liabilities, R4's accounting mode takes on a Ponzi-like intertemporal structure: in the current period *t* when AI takes over a certain workflow, an organization may immediately recognize cost savings, revenue growth, or performance improvements, and these organizational measures should not be equated directly with GDP or productivity gains; the pseudo-small-fibers (the true fiber is large—the system is going wrong—while the perceived fiber is small, with the interface still outputting normally; unfolded in §6), hidden technical debt (Sculley et al., 2015), and the de-skilling of human operators (Chapter 35 §2.3.5—the intensification of cognitive substitution, in which platforms extend the centralized processing of information distribution to the centralized processing of cognitive production itself) accumulated during the same period do not appear on the books as cognitive-technical debt; the subsequent maintenance, rework, and losses only become visible in some future period τ in the form of project failures, system rewrites, takeover incapacitation, and retroactive rework. The gap between current-period booking and future exposure has already been formalized in Chapter 22 as the phase transition of shadow debt $D_c^{shadow} = D_c^{true} - D_c^{obs}$ at the critical point of the manifestation function $M(\tau)$. The structure resembles Ponzi finance in "using new deployment to keep earlier liabilities from surfacing"; the chapter treats this as a risk mechanism, not as a characterization of all AI reinvestment. Throughout the entire duration of [t, τ], physical energy consumption is fully combusted: while a Ponzi scheme in the financial dimension can terminate with default and liquidation, in the physical dimension it cannot—the combustion has already occurred.

Reinforcing Loop R5: Infrastructure Lock-in. Build data centers → sunk costs → strengthened incentives to raise utilization → drive AI application deployment → need more data centers. The construction period of a single data center building itself (excluding preliminary site selection, permitting, and grid interconnection queuing) is typically 12–18 months (NERC, 2025a: *Characterization and Risks of Emerging Large Loads*); for large-scale (300–1000 MW) hyperscale projects, the grid interconnection application and queuing schedule itself has become a new determinant of cadence (U.S. DOE, 2024), with a design life of 15–25 years. Once built, operators have a strong economic incentive to fill them—idle servers are pure cost. This means that today's infrastructure decisions raise the energy-consumption baseline for the next ten to twenty years. When Microsoft, Google, and Amazon each announce tens of billions of dollars in data center construction plans, they are not just predicting future demand—they are creating future demand, because built facilities and long-term contracts create strong operating inertia—though this neither guarantees full utilization nor constitutes an immutable energy-consumption floor. This is a material path dependency: once concrete and steel and fiber optics are poured and laid, they become constraints on future decisions rather than outcomes.

The temporal rigidity of R5 is already beginning to counter-pressure demand from the supply side in the present: Chung et al. (2026) document that the interconnection queuing delays for newly built AI data centers have reached "multi-year" magnitudes, becoming a physical bottleneck for AI capacity expansion itself—this state of "approved for construction but unable to connect to the grid for a prolonged period" on the one hand pushes up the incentive for the utilization rate of already-built facilities (thus further reinforcing R5), and on the other hand pushes operators toward long-term contracts, co-located supply, or other procurement arrangements—which do not necessarily bypass the public grid (see the Three Mile Island case later in this section)—it may also delay or curtail some new capacity; this chapter counts its already-observed hoarding and diversion effects. This supply rigidity has received independent corroboration from industry research: JLL's *Global Data Center Outlook 2026* lists power—not location or cost—as the primary site selection criterion for data centers (due

to multiyear grid connection wait times), records that sites without secured power are increasingly stranded regardless of zoning or demand, and estimates that nearly 100 GW of new data centers will be added globally between 2026 and 2030 (doubling capacity), with combined real estate and tenant capital expenditures of up to $3 trillion by 2030 (JLL, 2026).

These four reinforcing loops and the Jevons mechanism are mutually nested. The demand generated by R1 is locked into physical infrastructure by R5; when rebound exceeds the savings, the Jevons mechanism converts efficiency improvements into an amplification of this lock-in; R3 accelerates the iteration speed of R1; R4 provides continuous capital input to the entire system and endows it with economic legitimacy. They are not five independent growth engines, but five faces of a single coupled system.

The balancing loops opposing them number three, but their strengths and time scales constitute an asymmetric configuration.

Balancing Loop B1: Energy Price Signals. Computing demand increases → electricity demand increases → electricity prices rise → computing costs rise → demand is dampened. This loop should theoretically provide negative feedback, but in practice it is weakened by multiple factors: technology companies sign long-term Power Purchase Agreements (PPAs) locking in low electricity prices; governments, out of strategic competition considerations, provide electricity subsidies or priority power supply guarantees to data centers; new nuclear and renewable energy projects are specifically built to meet AI demand. The last point is particularly noteworthy: when newly built clean energy is used to meet the incremental demand of AI rather than to displace the existing stock of fossil fuel generation, it is recorded as "green" in carbon accounting, but at the Earth-system level it does not reduce total emissions—it merely clothes the expansion of AI in the garb of renewable energy. The weakening of energy price signals means that the braking force of market mechanisms on AI expansion is far lower than textbook expectations. This structural change has been explicitly named in the power systems engineering literature: Zhang et al. (2026) diagnose hyperscale AI data centers as **microgrid prosumers** that possess both consumption capability and self-generation, energy storage, and bidirectional grid interaction capabilities, actively managing their own electricity procurement curves across multiple time scales through long-term PPAs, co-located clean generation, and self-built energy storage—electricity prices remain a dispatch variable in their model; this means that long-term contracts and self-supply change price exposure and response modes, and the applicability of the textbook "marginal consumer" simplification to the category of AI data centers is declining—not that price constraints themselves have failed.

The physical deceptiveness of the "net-zero" narrative has a further, deeper layer, involving the topological structure of the grid rather than merely the accounting standards of carbon accounting. AI training clusters—the parallel computation of thousands to tens of thousands of GPUs—demand uninterrupted, highly reliable power around the clock, while their power ramps up and down in synchronized steps at the second level—sustained high load coexists with rapid fluctuation. This load profile is structurally mismatched with the supply profile of renewable energy: solar power goes to zero at night; wind power fluctuates randomly with weather; neither can independently provide baseload. When a technology company announces that its data center is "100% powered by renewable energy," it usually means one of three situations: it has purchased an equivalent amount of Renewable Energy Certificates (RECs), which is a pure accounting operation that does not change the physical dispatch of the grid; it has signed a renewable energy PPA, but during periods when renewable energy is unavailable (night, calm, overcast), the gap must be filled by other sources, storage, or load adjustment, with emissions consequences depending on actual dispatch; or it has built accompanying energy storage systems, but

at currently disclosed deployment scales, the capacity and economic viability of battery storage remain far from sufficient to fully smooth baseload demand at the level of tens of thousands of GPUs.

In grids where additional clean supply is insufficient, storage and load flexibility are limited, and fossil marginal generation fills the gap, the following outcome can occur: the data center consumes solar power during the day that could have displaced fossil fuels (because without the data center, that solar power would have pushed fossil fuels off the grid), and at night forces the grid to fill its uninterruptible load gap with the marginal units of the time—typically natural gas peakers. The net effect: the increment of renewable energy is swallowed by AI, and the stock of fossil fuel generation is not displaced. The carbon accounting report shows "green," because the daytime electricity indeed came from solar. But the Earth's atmosphere does not read accounting reports—it only looks at total emissions, and in this scenario total emissions have not decreased, and annual renewable matching does not establish a reduction in total system emissions, because the nighttime combustion of fossil fuels offsets the daytime contribution of renewable energy.

This is not merely "greenwashing"—it is more structural than greenwashing. Greenwashing is deliberate deception. What occurs here may not involve any subjective malice: technology companies may sincerely believe they are doing the right thing; carbon accounting standards may have been reasonable at the time they were formulated. The problem is that the accounting framework was designed for a world where marginal changes could be treated as incremental adjustments, whereas AI's electricity demand—growing at rates of 15% to 30% per year—is no longer a marginal change; it is reshaping the dispatch logic and investment priorities of the grid itself. When the increment becomes so large that it alters the system structure, the accounting tools designed for increments can no longer track the real physical consequences they cause. The accounting is keeping pace; the physics is escaping. The gap between the two is the dwelling place of the "net-zero" pledge.

He Tingbo, President of Huawei's Semiconductor Business Department, pointed out publicly in her 2026 paper that the time-miniaturization acceleration principle of AI systems does not inherently contain an energy consumption constraint—"a super-node operating 10× faster but with 10× greater power consumption violates no scaling principle, yet exceeds grid capacity" (He, 2026). This is not an external critique from environmental organizations, but an open self-statement by a leading chip designer on the tension between its own roadmap and grid carrying capacity.

The most engineering-credible response to the baseload problem is nuclear fission, and it is also the direction in which AI infrastructure capital is currently placing the densest bets: Microsoft, on September 20, 2024, signed a **20-year** PPA with Constellation Energy, exclusively locking in the approximately **835 megawatts** of baseload power output from the restarted Three Mile Island Unit 1 (renamed Crane Clean Energy Center, in honor of the late nuclear industry leader Chris Crane), planned, at the time of the 2024 announcement, to come back online in **2028**; this PPA is the largest in Constellation's history and comes with an independent process to apply to the U.S. Nuclear Regulatory Commission (NRC) for extending the operating license to 2054 (Constellation Energy, 2024-09-20); Constellation subsequently announced in 2025 that the restart had been accelerated to **2027** (Constellation Energy, 2025-09); Google, on October 14, 2024, signed the world's first corporate multi-unit SMR PPA with Small Modular Reactor (SMR) developer Kairos Power, with a total scale of approximately **500 megawatts**—the first SMR planned to come online by **2030**, with subsequent deployments continuing to **2035**; this agreement covers Kairos's subsequent commercial SMR fleet, distinct from its Hermes non-power test reactor in Oak Ridge, Tennessee, which has already received NRC construction permit (Google, 2024-10-14; Kairos Power, 2024-10-14); Amazon, in October 2024, partnered with Dominion Energy to explore an SMR

project next to the North Anna nuclear power station, and anchored a roughly $500 million financing round in X-energy (Amazon, 2024-10-16; X-Energy Reactor Company, 2024-10-16); and Sam Altman, CEO of OpenAI, previously served as Chairman of the board of the advanced nuclear fission company Oklo, stepping down in April 2025—in the announcement, Altman himself juxtaposed the timing of his resignation with the company's direction: "As Oklo explores strategic partnerships to deploy clean energy at scale, particularly to enable the deployment of AI, I believe now is the right time for me to step down"; the company stated it would "continue to explore strategic partnerships with leading AI companies, including potentially with OpenAI" (Oklo, 2025).

The superiority of nuclear power on the carbon dimension must be acknowledged: its operational-phase carbon emissions are near zero, and its full-lifecycle carbon emissions are on the same order of magnitude as wind power. If this chapter were to avoid this point, it would lose its honesty.

But after acknowledging the improvement on the carbon dimension, the physical cost of this improvement must be interrogated. Nuclear fission does not eliminate waste heat—it changes the disposal pathway of waste heat. The thermal efficiency of a pressurized water reactor nuclear power plant is approximately 30%, meaning that for every unit of electricity generated, over two units of heat-equivalent waste heat must be discharged. By comparison, a natural gas combined cycle power plant has a thermal efficiency of approximately 55% to 60%. This means that, for every unit of electricity generated, a nuclear plant needs to discharge close to three times the waste heat of a natural gas combined cycle plant; whether the corresponding evaporative water consumption increases in the same proportion depends on the cooling method—once-through cooling is dominated by warmed discharge, while cooling towers are dominated by evaporation. According to the engineering benchmark given in the U.S. DOE (2006) report *Energy Demands on Water Resources*, a 1 GW-class pressurized water reactor nuclear plant operating at full capacity for one year evaporates water through cooling towers typically on the order of tens of billions of liters; as a real-world case reference, the Calvert Cliffs Nuclear Power Plant in Maryland uses once-through cooling from Chesapeake Bay, withdrawing approximately 3.4 billion gallons (approximately 12.8 billion liters) of cooling water daily from the bay in 2022, while the report estimates daily evaporative losses associated with the heated discharge at approximately 18.4 million gallons—about 0.5% of the withdrawal, with the vast majority returning to the bay as heated discharge (Maryland DNR, 2024: *Surface Water and Groundwater*).

Here there is a stacked structure worth naming; this chapter terms it **double evaporation**: where both the electricity supplier and the data center use evaporative cooling, generation-side and on-site consumption should be counted together; whether they affect the same watershed depends on the location of the power plant, the water sources, and the supply arrangement—where the supply side uses once-through cooling, major impacts involve large withdrawals and heated discharge, while evaporative losses can still occur (as the Calvert Cliffs case above shows). Its cumulative effect exhibits threshold-triggering characteristics in regions already under water stress: when net consumption persistently exceeds the natural recharge rate of an aquifer or watershed, storage will keep declining; whether the system jumps from a sustainable state to a degraded one depends on the specific hydrological and ecological thresholds.

Thus, the switch from fossil fuels to nuclear power performs, in terms of cost structure, a double redirection. The first is a spatial redirection: CO_2 is a long-lived, well-mixed greenhouse gas, its consequences distributed throughout the entire atmospheric system; water is an extremely regionalized resource, the consequences of its depletion concentrated in specific watersheds, specific aquifers, specific communities. When a technology company writes "zero-carbon electricity" on its carbon accounting report, what physically occurs is a transfer

of costs from the global atmosphere to the water systems in the localities of the data center and the nuclear plant. The second is a temporal redirection: the climate consequences of carbon emissions manifest through decades of cumulative delay—this is precisely why the fossil fuel economy could operate for two centuries without "hitting the wall"; the depletion of water can be measured annually, observed quarterly, perceived monthly. For nuclear–data-center combinations using consumptive cooling in water-stressed basins, low-carbon supply may coincide with more concentrated local water pressure; which constraint becomes binding first—climate-related constraints, or a series of regional water crises on a nearer time scale—depends on site, water source, cooling design, and climate: a city's reservoir running dry, a river running dry in summer, an agricultural district's irrigation quota being cut to prioritize "critical infrastructure"—and the power to define critical infrastructure, once again, belongs to capital.

Finally, the additionality problem—far from disappearing in the nuclear scenario, it becomes even sharper. Microsoft has locked in the entire 835 MW output of the restarted Three Mile Island unit through a 20-year power purchase agreement; the electricity still flows into the PJM interconnection (the largest regional transmission organization in the United States), and what Microsoft purchases is its energy attributes and financial commitment. Without this agreement, the unit, by the operator's own account, would not be restarted—evaluating this procurement agreement requires comparison with an equivalent-load scenario without that agreement; evaluating additional AI deployment requires a separate baseline without the additional load—the fact that "this carbon-free capacity exists because of AI demand" corresponds to the latter. The two comparisons may yield different answers and are not interchangeable. Microsoft's agreement currently supports its carbon-free matching goal; whether that goal corresponds to a real reduction in system emissions must be judged against both baselines separately. A procurement agreement may deliver genuine emissions reductions relative to an equivalent-load baseline even while additional AI deployment increases emissions relative to a no-additional-load baseline; procurement additionality and the overall consequences of deployment must therefore be assessed separately. Failure to achieve an absolute reduction relative to the no-additional-load baseline does not, by itself, demonstrate a failure of procurement additionality.

Ultimately, the role nuclear power plays in system dynamics is not as a component of the balancing loop B3, but as an enabling condition for the reinforcing loop R1. It does not slow the expansion of AI; it removes a physical bottleneck on the expansion path—or, more precisely, it may, locally, allow the water constraint (a regional, immediate, physically hard-to-negotiate constraint) to become binding before the carbon constraint (a global, delayed, politically manageable one)—which comes first likewise depends on site and cooling design. Jevons Paradox reappears, but in a more concealed form: carbon-intensity improvements may be partially or wholly offset by scale expansion—more data centers, more GPUs, more training runs can absorb the efficiency gains—and carbon, water, and other resource consequences should be accounted for separately rather than presumed never to decline in aggregate. And in the cases examined in this chapter, the water constraint appears in a more regionalized form, with impacts that are more directly felt than the carbon constraint.

This escape impulse no longer needs speculation. It is surfacing in the form of concrete engineering plans and capital deployments. The ultimate imagined magnitude of AI infrastructure's energy demand has been pushed to the level of a Dyson Sphere—a hypothetical megastructure capturing the output of a star to serve a planetary civilization—this chapter treats it as a far-horizon reference for the expansionary imaginary, not as a validated demand forecast. Caltech's space solar power project has already completed a proof-of-principle demonstration of microwave power transmission in orbit (Caltech, 2023). Moving heavy industry—including

computation—off the surface of the Earth is Bezos's publicly stated position (Clark, 2019); Musk's program is to make humanity a multi-planetary species (Musk, 2017)—the two share the structural direction of moving human activity off the surface of the Earth. The importance of these signals lies not in whether space-based data centers will be realized in ten or twenty years—most likely they will not—but in the deep logic they reveal about the relationship between capital and planetary boundaries.

Faced with the physical constraints described in this chapter—the baseload bottleneck of the grid, the regional depletion of water resources, the structural gap between carbon accounting and physical emissions—capital's first reaction is not contraction, not a reassessment of the premises of growth, but a search for new physical space in which to continue growth. This is completely consistent with the historical pattern of encounters with resource boundaries: fifteenth-century Europe, faced with the limits of local resources, did not choose to reduce material consumption but chose transatlantic colonization; nineteenth-century industrial capital, faced with the limits of local labor and raw materials, did not choose to slow industrialization but chose imperial expansion. The space escape narrative is the contemporary variant of the same structural impulse: not turning back in the face of constraints, but attempting to dismantle the constraints themselves.

But the very attempt to dismantle is itself an acknowledgment of the reality of the constraints. When a company begins seriously calculating the launch costs of putting GPUs into orbit, it has already, at the behavioral level, admitted a fact it still denies at the PR level: the Earth's physical capacity is becoming a constraint on AI expansion, and this constraint cannot be dissolved merely by "net-zero" pledges and carbon credits. The engineering feasibility of space-based data centers is an engineering question. Their diagnostic value as a symptom is a deeper question—it tells us that the growth impulse driving the Great Acceleration has become powerful enough that it is preparing to breach the atmosphere to sustain itself. Any policy narrative that claims this impulse can be "tamed" through voluntary commitments, carbon taxes, or environmental, social, and governance (ESG) ratings needs to explain: if these mechanisms are effective, why are the companies most capable of complying with them planning to escape the physical domain they constrain?

Balancing Loop B2: Regulatory Constraints. Environmental impacts become visible → public concern → regulatory intervention → cost internalization → expansion speed reduced. The delay time of this loop is extremely long—from the visibility of impact to the effectiveness of regulation typically takes years to decades—and it is continuously weakened by intensive lobbying and the narrative of "strategic competition." In the current geopolitical landscape, any environmental regulation that might slow the speed of domestic AI development is immediately framed as "letting the adversary win." The U.S.-China competition narrative plays an ironic function here: it grants both sides' AI expansions the legitimacy of national security, thereby making environmental regulation the issue with the lowest political capital among all AI governance topics.

This diagnosis is not speculative but has a directly referable precedent path along U.S. military history. Crawford (2022) documents an institutional fact nearly forgotten in public discussion outside climate networks: **the Kyoto Protocol (1997), during the signing negotiation stage, established exemptions excluding certain military greenhouse gas emissions from national totals (such operations still being reported as memo items)**. The negotiating team, led by the U.S. Department of Defense, explicitly blocked any "full and transparent accounting of . . . military greenhouse gas emissions" from entering the international mechanism, on the grounds that the U.S. military wanted to maintain military preeminence—which its leaders believed required the capacity to have unconstrained greenhouse gas emissions; after the Paris Agreement (2015), full disclosure and military emissions cuts were left as voluntary matters (optional), and as of Crawford's (2022) account, the U.S.

had still not fulfilled comprehensive disclosure. Even more noteworthy is the language embedded in the Biden administration's 2021 whole-of-government net-zero directive: the Department of Defense and other agencies could be exempted "when it is in the interest of national security" (Crawford, 2022). An environmental exemption pathway articulated from a national security perspective has therefore run for decades since 1997—a full climate negotiation cycle, institutionally verified as a stable structure.

And AI infrastructure possesses the structural conditions to be stitched into this system along the same historical trajectory: at the current moment when the U.S.-China strategic competition narrative is continuously intensified, AI's physical footprint is being increasingly frequently reclassified under the rubric of "participating in national competition." Crawford's deep cycle diagnosis provides a structural warning here: the long-cycle feedback among economic growth—fossil energy use—military mobilization—geographic expansion over the past 170 years left military emissions without comprehensive accountability—incomplete sectoral disaggregation, exceptions for particular operations, and supply-chain disclosure gaps; when AI infrastructure is endowed with the same status of "unconstrained capability" within the U.S.-China strategic competition narrative, it possesses the institutional conditions to exit effective environmental accounting along the same pathway.

Balancing Loop B3: Planetary Boundary Hard Constraints. Cumulative impact → boundaries breached, risk keeps rising → resource and ecological constraints tighten → expansion is curbed (crossing a tipping point is one steeper risk path). This is the ultimate balancing loop—but it is worse than "catastrophic": it is unequally catastrophic.

The implicit premise of B3 is: when physical constraints tighten, AI infrastructure will bear the shock together with other social systems. But this premise ignores the resilience privilege of capital. When a region's power grid nears collapse under extreme heat, the users with the highest payment capacity and strongest political negotiating power—data centers—are, by this ordering, expected to sit at the end of the curtailment sequence; this is a structural prediction, not yet confirmed by systematic outage records. Before that, residential electricity will be restricted, agricultural irrigation will be reduced, manufacturing will be ordered to stagger peak loads. During the 2021 Texas winter storm, the grid lost 34 GW of generation capacity over two consecutive days and was forced to shed 10–20 GW of firm load; nearly 70 percent of Texans experienced outages at some point between February 14 and 20, and at least 210 people died (FERC, 2021; Texas Comptroller, 2021)—it was the residential distribution side that bore the shock of the load shedding. When water resources approach their limit, the adaptation strategies of technology giants—desalination, inter-basin water transfer agreements, relocation to higher-latitude regions with relatively abundant water resources—are technically feasible; in the absence of institutional constraints protecting other water users, their actual exercise may shift a substantial part of the cost of scarcity onto subjects with weaker adaptive capacity—surrounding farmers, communities dependent on the same aquifer, downstream residents ceding water rights for transfer projects.

If this priority in resource access materializes during a crisis, the dynamics of the B3 loop are not "environmental collapse → AI deceleration" but "environmental collapse → intensified resource competition → AI maintains operation through payment capacity advantage → the cost of collapse is shifted onto vulnerable groups outside the system → AI continues to operate until a physical limit that even the capital moat cannot withstand is reached." In this revised dynamics, B3 as a balancing loop targeting AI expansion may see its constraint significantly postponed, falling earlier on groups with weaker adaptive capacity—it is overridden by a stronger mechanism (capital's priority claim on resources). B3 is indeed a balancing loop, but what it balances is not AI, but the human society outside the AI system. The hard constraint of planetary boundaries will eventually halt

this train—no capital can defy the laws of thermodynamics—but before it finally halts, this train will first run over everyone outside the carriages.

This constitutes a double temporal injustice. AI's physical cost is already spatially unjust—combustion occurs at the data center; victimization occurs in climate-vulnerable regions. The unequal triggering mechanism of B3 introduces a further injustice in the time dimension: AI infrastructure will survive environmental collapse longer than the communities it harms. Victims disappear before the harming system does—this eliminates the feedback signal. When farmers in arid regions have already lost their water source, the data center is still running—and the very act of running means the signal "everything is normal" is still being sent. The system cannot perceive the damage it has already caused, because it has already eliminated the sentinels that could have raised the alarm.

System dynamics' diagnosis of this structure is straightforward: when the cadence of the main reinforcing loops outpaces the response of the balancing loops—quarterly-to-annually against annually-to-decennially—and R5's multi-year lock-in deposits the amplification into stock, the system faces a risk of overshoot—it may first greatly exceed the sustainable level, then contract violently in the face of some hard constraint; this is a risk scenario to be tested, not an established outcome. Donella Meadows (2008) expressed this concisely: in a negative feedback system with delays, the stronger the growth loops, the more severe the overshoot, the more violent the collapse.

§4 Planetary Boundaries

In 2009, Johan Rockström et al. proposed the "Planetary Boundaries" framework, identifying nine Earth-system processes with safe operating boundaries (Rockström et al., 2009). The core insight of this framework is not that "Earth's resources are finite"—that is common knowledge—but that the Earth system has nonlinear tipping points, and that planetary boundaries are precautionary limits set to safeguard Earth-system stability and resilience, which do not all correspond to individually identifiable tipping points: transgression increases risk; some of these processes and their interactions may trigger abrupt shifts, or changes that are difficult to reverse on human timescales. The 2023 assessment by Richardson et al. found that six of the nine boundaries had already been breached (Richardson et al., 2023).

The *Planetary Health Check 2025*, published by the Planetary Boundaries Science (PBScience) project led by the Potsdam Institute for Climate Impact Research (PIK), updated this figure to seven boundaries breached—ocean acidification was assessed as having crossed the boundary for the first time in the 2025 report (PBScience, 2025). Only two boundaries remain within the safe operating space: atmospheric aerosol loading (improving globally) and stratospheric ozone depletion (remaining stable). The Earth system is already outside the safe operating space of the Holocene. PHC 2025 simultaneously cites orbital dynamics calibration: if too many irreversible tipping points can be avoided, this class of Holocene stable state could have persisted for another approximately 50,000 years before entering the next glacial period (PBScience, 2025: §2.1); the current transgression does not diminish the Earth itself on the Earth's time scale, but diminishes, on civilization's time scale, the possibility that "the Holocene epoch, during which agriculture and modern civilizations developed" remains open to humanity.

AI's physical footprint directly intersects with at least three of these boundaries.

The Climate Change Boundary. This is the most direct intersection. Carbon emissions from AI data centers

directly contribute to the rise in atmospheric CO_2 concentration. According to the estimate given by IEA's *Energy and AI* (2025), data centers accounted for approximately 1.5% of global electricity consumption in 2024 (approximately 415 TWh), with corresponding carbon emissions on the order of hundreds of millions of tonnes of carbon dioxide (CO_2)—the IEA estimates emissions from data center electricity use at roughly 180 Mt CO_2 today, reaching approximately 300 Mt by 2035 in the Base Case (IEA, 2025).

But the crucial difference lies in the trajectory: the carbon emission growth of the aviation industry is roughly synchronous with global economic growth—international aviation's greenhouse gas emissions grew at an average annual rate of approximately 3.4% between 2010 and 2019 (IPCC, 2022)—while data center electricity consumption is growing at a much faster rate (a strict comparison of emissions against emissions would also need to account for grid carbon intensity)—the IEA's baseline in *Energy and AI* is an average annual global data center electricity consumption growth of approximately **15%** from 2024 to 2030 (IEA, 2025); the actual growth rate in 2025 already reached **17%**, with AI-focused data centers growing at approximately **50%** in 2025 (IEA, 2026). Goldman Sachs (2025) predicts that global data center power demand will grow by approximately 165% by 2030 compared to 2023 levels (Goldman Sachs, 2025). If the initial scales are the same, growing at 3% and at 15% annually for ten years, the latter will be roughly 3 times the former ($1.15^{10}/1.03^{10} \approx 3.0$); for AI-focused data centers growing at 50%, roughly 43 times ($1.5^{10}/1.03^{10} \approx 42.9$). These compound-growth examples illustrate the effect of fixed growth rates on long-run magnitudes, not an emissions forecast; the IEA figures apply to electricity-related emissions from all data centers, an AI-specific emissions trajectory additionally requires scenarios for AI's electricity share and supply carbon intensity.

How AI-specific emissions pressure on the remaining carbon budget will evolve depends on multiple uncertain parameters (deployment speed, grid decarbonization pace, efficiency improvements); the chapter identifies this as a risk to be tracked rather than a determined emissions trajectory.

The Freshwater Change Boundary. Following the 2022 proposal and assessment of the green-water boundary, the 2023 update incorporated "green water" (soil moisture) into the formal assessment of the freshwater boundary and judged this boundary to have been breached. The water consumption of AI data centers falls under the category of "blue water" (surface water and groundwater); the evidence reviewed here does not quantify AI's contribution to global freshwater-boundary transgression; the share of global water consumption cannot substitute for an assessment of pressure in particular watersheds—this global judgment obscures regional realities. Data center site selection often prioritizes electricity cost and network latency over water resource conditions. When a data center consuming several billion liters of water annually is built in the already water-stressed American West or India, its impact on the local water system can be severe. This geographic mismatch—AI's beneficiaries are dispersed across the global consumer side, while its water resource costs are concentrated in specific watersheds—constitutes a spatial externality transfer that cannot be adequately represented by global water-consumption totals alone. This problem needs to be tracked at the watershed scale, and watershed-scale data is precisely what is most lacking.

The Novel Entities Boundary. The "novel entities" boundary covers entirely new chemical substances and materials introduced into the biosphere by human activities—this boundary was judged breached in the 2022 assessment (Persson et al., 2022). The full lifecycle of AI hardware involves material flows at three levels. Upstream: rare earth element mining (neodymium, dysprosium, terbium, etc., used for high-performance magnets), lithium and cobalt mining (used for backup battery systems), ultrapure water and specialty chemicals (used for wafer fabrication). Midstream: perfluorinated compound (PFC) and other greenhouse gas emissions from the

semiconductor manufacturing process, the use of photoresists and etching gases. Downstream: the electronic waste generated by the large-scale hardware turnover flow driven by roughly biennial generational GPU updates (for which actual retirement ages lack public data), containing composite materials and toxic substances that are difficult to recycle. The growth rates of these material flows are coupled with the growth rate of AI deployment, and the latter—as analyzed in §3—is driven by positive feedback loops, with whether its growth is superlinear depending on the actual strength of the loops. The material footprint of the AI industry is its most invisible externality, because it is hidden at the farthest reaches of the global supply chain, and the environmental and health costs of many links are erased from the final product. In the user's eyes, AI is purely digital—text and images appear and dissolve on the screen. Its materiality is thoroughly concealed by the length and complexity of the supply chain.

The common structure of these three intersections is: AI's physical impact is not imposed on a blank Earth system but superimposed on an Earth system already under critical stress. This "superimposition" structure is crucial—it means AI's marginal impact cannot be assessed in isolation. In a system far from a tipping point, adding a new stressor has an impact roughly equal to the strength of that stressor itself. But in a system already near a tipping point, the impact of a marginal increment is highly nonlinear—the effect of the last straw may equal the sum of all previous straws. The figure that emissions from data center electricity use remain below 1.5% of total energy sector emissions (IEA, 2025) appears manageable in linear thinking. But in an Earth system where seven planetary boundaries have already been breached (PBScience, 2025; Richardson et al., 2023) and the carbon budget is nearly exhausted, an increment growing at an IEA baseline of **15%** annually, **17%** actual in 2025, with AI-focused data centers growing several times faster (IEA, 2025; IEA, 2026), is not a minor problem, but a driving force that could alter the system's trajectory within the coming decade.

§5 The Second Derivative: Recursive Acceleration and Its Constraints

If the preceding analysis describes the growth of AI's physical footprint—a positive first derivative—this section examines a further risk mechanism: the conditions under which recursive research feedback could raise the relative growth rate of total energy use (not merely a positive second derivative—exponential growth at a constant rate also has a positive second derivative).

The "Great Acceleration" is a core concept in Earth system science, used to describe the synchronous, dramatic rise of key indicators of human activity since the mid-twentieth century. Steffen et al. (2015) formalized this concept within Earth system science: one set of charts tracks twelve socioeconomic indicators from 1750 to 2010 (population, real GDP, foreign direct investment, urban population, primary energy use, fertilizer consumption, number of large dams, water use, paper production, transportation, telecommunications, international tourism); another set tracks twelve Earth system indicators over the same period (atmospheric CO_2, N_2O, CH_4 concentrations, stratospheric ozone, surface temperature, ocean acidification, marine fish capture, shrimp aquaculture, coastal nitrogen flux, tropical forest loss, terrestrial biosphere degradation, domesticated land); taken together, the graphs show a marked acceleration in human activity and Earth-system impacts after the mid-twentieth century, but their trajectories differ—some indicators later slow, stabilize, or show recovery; the Great Acceleration describes an aggregate historical transition, not a single exponential law shared by all twenty-four indicators (Steffen et al., 2015).

The paper simultaneously argues: among the candidate stratigraphic markers for the onset of the Anthropocene,

the mid-twentieth century, when the "Great Acceleration" began, is the most persuasive from an Earth system science perspective, because only after crossing the 1950 node did the state and functioning of the Earth system undergo a fundamental shift that exceeded the range of Holocene natural variability and was driven by human activities. McNeill and Engelke (2014), from the perspective of environmental history, provide a long narrative chronicle of the same period—population, energy, urbanization, consumption structures, biosphere changes after 1945—restoring the formalized curve cluster of Steffen et al. into "an environmental history of the Anthropocene." The Great Acceleration is not the story of a single technology or a single nation; it is the aggregate portrait of industrial civilization as a whole's systematic impact on the Earth system.

AI is now introducing a new source of acceleration into this already accelerating system. And this source of acceleration possesses a structural feature that no previous source of acceleration ever had: self-acceleration.

In the history of the Great Acceleration, every major acceleration driver—fossil energy, synthetic fertilizers, plastics, semiconductors—drove exponential growth in output through positive feedback loops of technology and economics. But the iteration speed of these loops was always limited by a common bottleneck: the time scale of human cognition and organizational capacity. The improvement of the steam engine required teams of engineers spending years on experimentation and trial-and-error. The optimization of the internal combustion engine depended on physical experiments and accumulated experience; every step of improvement required human engineers to conceive, test, and implement. Even the semiconductor industry's Moore's Law—the most sustained and rapid exponential improvement in the history of human technology—presupposed the continuous investment of thousands of engineers, with each generation of process node advancement requiring years of R&D and tens of billions of dollars of investment. In all these cases, humans were the rate-limiting factor in the loop—not because humans are not intelligent enough, but because the per-unit-time output of human time, attention, coordination capacity, and creativity is finite.

This judgment is not a discovery of the present era—von Neumann had already written its classic formulation in 1955: increased speed "did not so much shorten time requirements of processes as extend the areas of the earth affected by them," because "most time scales are fixed by human reaction times, habits, and other physiological and psychological factors" (Von Neumann, 1955).

AI is widening, even eliminating, this bottleneck. When AI is used to assist AI research—from code generation to literature review to experimental design to hyperparameter search to architecture exploration—the human bottleneck in reinforcing loop R3 is widened. This means the gain of the R3 loop is no longer a constant, but may become an increasing function of R3's own output: stronger AI → more efficient AI R&D assistance → faster production of even stronger AI → loop gain further increases.

In the language of system dynamics, this is a structural change from fixed-gain feedback to variable-gain feedback. In a minimal growth model, a constant positive net relative growth rate produces exponential growth—already fast enough, as exponential growth is the underlying dynamic of most resource depletion and environmental crises. If that rate rises over time, the super-exponential behavior considered here may emerge. The argument concerns an increasing-gain scenario, not variable-gain systems in general. This behavior is mathematically termed super-exponential growth; only super-exponential growth satisfying specific growth conditions (such as $X'=kX^p$ with $p>1$ in the absence of external constraints) has its state variable tending toward divergence in finite time (finite-time singularity). Of course, a physical system will not actually reach infinity—it will stop in the face of some hard constraint. But there are two ways to "stop": either decelerate in advance

(soft landing), or crash after hitting the boundary (hard landing). And under the current loop structure, the B1 and B2 loops that provide the force for advance deceleration are being systematically weakened.

Has AI's energy consumption trajectory already exhibited super-exponential characteristics? A precise answer to this question requires fine-grained time-series data currently unavailable to the public—this is itself a consequence of the data opacity problem diagnosed in §2. But the structural argument does not depend on precise data: R3 identifies a mechanism through which research may accelerate recursively—the available evidence is consistent with the widespread adoption of R3 in accelerating assisted R&D—but whether this raises the relative growth rate of total energy use also depends on deployment demand, energy per computation, investment responses, and supply constraints; under the condition that the loop gain itself increases sufficiently rapidly with the state variable, this structure points toward super-exponential behavior. The section therefore identifies a risk mechanism rather than an established energy trajectory. If such acceleration develops and persists, the constraints it encounters and the manner of its deceleration remain questions for further investigation.

This is the true meaning of the "Great Acceleration" concept for AI. AI is not merely a product of the Great Acceleration era, not merely a new curve in the Great Acceleration curve cluster. It is the first driving force in the Great Acceleration with recursive self-acceleration capability within the cognitive task domain—the first technology capable of accelerating its own acceleration. If the Great Acceleration is a train continuously picking up speed, every previous technology—fossil fuels, fertilizers, plastics, semiconductors—added horsepower to the train; AI is the first technology capable of climbing into the cab itself and continuously pressing down on the throttle. It is not just speed; it is the speed of speed.

Here, engagement is required with a counterargument almost invariably raised in this context: AI can also be used to address climate change—optimize energy grids, improve climate models, accelerate new material R&D, enhance carbon capture efficiency. This argument is valid at the level of individual applications and can even be quite impressive in specific cases. But placed at the level of system dynamics, individual positive applications do not by themselves establish net emission reductions at the system level; absent consistent counterfactuals and lifecycle accounting, "AI for climate" can also serve as a legitimizing narrative for expansion—using more AI to solve problems that AI itself participates in creating, while the physical footprint of "more AI" itself exacerbates that very problem. Unless the emission-reduction benefits of "AI for climate" are rigorously accounted for as net benefits—i.e., remain positive after subtracting the full-lifecycle carbon footprint of that AI—"AI for climate" is merely a rhetorical device that makes expansion self-justifying. At present, per-application net-benefit accounting for "AI for climate" remains scarce—existing studies are mostly scenario estimates and reviews (see below), with no consistent counterfactual baseline.

A study at the corporate governance level provides negative micro-level evidence. Xie Tingting and Fang Xinyi (2026), on a sample of Chinese listed firms from 2010–2024, found that enterprises' deepening use of AI technology does not net-mitigate climate risk, but instead amplifies their climate risk exposure through four organizational governance mechanisms—the mechanics of which, as read in this chapter, are: increasing the degree of business diversification (broadening business radius exposes the enterprise to stacked shocks from more types of events such as ice and snow, floods, extreme heat events simultaneously), attracting analyst attention (after market pricing benchmarks are raised, the reflexive price of low-probability tail events is amplified), strengthening executive risk appetite (algorithm-assisted decision-making biases executives toward accepting higher unhedged exposures), and intensifying financing constraints (micro-level compression of enterprises' capacity to reserve capital redundancy for climate shocks); heterogeneity analysis indicates that this amplification

effect is more pronounced in state-owned and non-high-tech industry enterprises (Xie Tingting & Fang Xinyi, 2026). This result is only micro-level evidence from a Chinese sample and cannot be directly extrapolated into a global diagnosis; however, its function in the argument of this section is negative—even with only one such quantifiable micro-example existing, the narrative of automatic risk mitigation can no longer be taken as the null hypothesis at the empirical level; as for the net emissions effect of AI applications, this study does not directly test it. These amplification pathways emerge as enterprises adopt intelligentization, and their strength is moderated by firm characteristics such as R&D investment—they are not accidental phenomena of "misusing AI"; this and the four reinforcing loops and Jevons mechanism described in §3 are not the same input port, but independently observable evidence of the same system state at the organizational level.

More critically: with explicit system boundaries and counterfactuals, net energy use and emissions can in principle be estimated, though with substantial uncertainty; what such accounting cannot by itself determine is a deployment's use value, distributional justice, and ethical legitimacy—a unit of electricity saved by using AI to optimize the grid and a unit of electricity combusted to train a large model are the same unit of electricity in thermodynamics—the physical quantities remain comparable—though they support different kinds of uses. This point will be made precise in §6.

At scales larger than corporate governance, three further studies provide complementary evidence on resource footprints and deployment scenarios; they do not test the same outcome variable as Xie Tingting and Fang Xinyi (2026). Xiao et al. (2025) conducted scenario-based estimates of the energy–water–climate nexus impact of U.S. AI server deployment from 2024–2030: within the U.S. alone, AI servers could generate an annual water footprint of **731–1,125 million cubic meters**, with additional annual emissions between **24–44 Mt CO_2 equivalent**; best practices could theoretically reduce carbon and water footprints by **73%** and **86%**, respectively, but their effectiveness is "constrained by current energy infrastructure"—a physical lock-in gap diagnosed by §3 R5 exists between theoretical and achievable reduction magnitudes (Xiao et al., 2025).

Turkay et al. (2026) extend the analytical scale to global and 2050: within a scenario framework, even under scenarios with significant improvements in hardware and algorithmic efficiency, total global AI electricity demand by 2050 will still grow by **more than sixfold** compared to 2024 levels; under a business-as-usual scenario, AI electricity consumption could reach **30%** of global electricity demand, corresponding to annual emissions of **>8 Gt CO_2 equivalent** (Turkay et al., 2026). This global order-of-magnitude figure is directionally consistent with §3 R2 (the Jevons mechanism)—but it is the output of scenario projection, not direct empirical corroboration of the loop; per-unit efficiency improvements do occur with each hardware generation, and aggregate growth outpacing them is its scenario result. Wheeler et al. (2026), in a systematic review of 73 peer-reviewed empirical studies from 2021–2025, unfold the temporal structure into a three-stage "synthesized S-curve" heuristic—the early deployment phase of AI (0–2 years) shows emission reductions; the middle phase (2–5 years) sees rebound effects; the long-term (5+ years) decarbonization benefit depends on conditions: regions with high renewable energy shares and water abundance can reach net environmental benefits within 2–3 years, while fossil-fuel-dependent, water-stressed regions "may never reach net positive outcomes"—the review records geographic variation reaching **10–60 times** (Wheeler et al., 2026).

This paragraph only borrows the findings layer of these three sets of studies, without also borrowing their respective normative claims (e.g., policy recommendations concerning coordinating private actors with regulatory intervention, building renewables before deployment, etc.), which belong to the solution layer and are a different question from the "triple collapse of existing pricing tools" diagnosed in §7 of this chapter. The three studies

provide a regional scenario estimate, a global long-term projection, and a synthesis across studies; together they point to the importance of scale, location, energy systems, and rebound effects—but they cannot be merged into direct evidence of a continuously increasing growth rate of total AI energy use, nor proof that the acceleration mechanism discussed in §5 has been observed.

Here, a reverse slippage of argument must be simultaneously ruled out. The thesis of this chapter is **not** "to protect the environment, we should reduce the use of AI," nor does it substitute a uniform increase or reduction in usage for application-specific judgment. Advocating that a particular use be reduced may rest on that use having no commensurate value, or on its costs outweighing its value—neither requires accepting the premise that "AI usage = productivity." What this chapter seeks to establish is a question one step prior: when the ratio of "genuine productivity vs. pseudo-productivity at the same energy consumption" cannot be accounted for ex ante, neither "always reduce" nor "always increase" is at the correct problem level; what is needed is identification of the benefits, costs, and alternatives of each specific use—what truly needs inventorying is which combustion purchases genuine output, and which merely accelerates the burning of the Earth to produce hallucinations, sycophancy, and other pseudo-productivity.

Weizenbaum (1976), in Chapter 10 of *Computer Power and Human Reason*, gave a two-part criterion: when a computer application simultaneously satisfies two conditions—side effects that are "irreversible and not entirely foreseeable," and that "cannot be shown to meet a pressing human need that cannot readily be met in any other way"—it "ought not to be pursued." This is a **conjunctive** criterion, not two slogans from which one can pick and choose.

Weizenbaum's criterion is directed not at the computer itself, but at a deployment distortion—placing the machine in a position where it ought not to substitute for humans (judging guilt on behalf of humans, performing psychotherapy on behalf of humans, "understanding" on behalf of humans), while simultaneously allowing humans to outsource their responsibility for their own judgment; he explicitly states that his ground for rejecting such deployments is not technical infeasibility but immorality—"It has nothing to do with what computers can or cannot be made to do" (Weizenbaum, 1976). He was himself the author of ELIZA and witnessed firsthand the moment when this placement was taken seriously as "conversation"; what he sought to safeguard was not "use computers less," but **keeping the computer in the position where it can stand**—and where to place it, what to dress it up as, has always been a human decision, not an attribute of the machine.

The same discernment was articulated even more sharply by Dijkstra. In his 1984 lecture in Austin, Texas (EWD898, "The Threats to Computing Science"), he criticized both of the founding figures, Turing and von Neumann: Turing posed the question as "whether Machines Can Think"—"a question of which we now know that it is about as relevant as the question of whether Submarines Can Swim"; von Neumann, for his part, drew "analogies sufficiently wild to be worthy of a medieval thinker" between computers and the human brain (Dijkstra, 1984). What these two parallel identifications point to is not a later corruption, but a category error already embedded in the coordinate system of the problem at the founding of the discipline. A submarine is not a fish, but a submarine can travel farther in the water than a fish—on the condition that it is not forced to "swim like a fish." A computer is not a brain, but it can do things in symbolic space that a brain cannot—on the condition that it is not packaged as another brain.

As early as EWD498 ("How Do We Tell Truths that Might Hurt," 1975), Dijkstra put this even more directly: treating computing systems in anthropomorphic terms is "a symptom of professional immaturity"; and the tools

we use "have a profound (and devious!) influence on our thinking habits, and, therefore, on our thinking abilities" (Dijkstra, 1975).

Placing the discernments of these two figures within the argument sequence of this chapter: the physical footprint argued in §§3–5—concerning irreversible carbon emissions, irreversible biosphere degradation, hydrological jumps crossing tipping points—falls into the first half of Weizenbaum's criterion (irreversible and not entirely foreseeable). But **the first half alone is not sufficient to conclude "ought not to be pursued"**—Weizenbaum's criterion is conjunctive, not disjunctive. The second half (whether this combustion has been exchanged for a pressing human need that cannot readily be met in any other way) is not resolved in this chapter; it requires not a section but an entire book's argument. This chapter makes no judgment on the latter condition for now, points out that it must be asked, and points out that it **cannot be answered by these accounting metrics alone**—this is precisely what §§6 and 7 proceed to address.

This suspension is not a rhetorical concession. It points toward a more precise question: **the problem of burning the Earth is not just "burning too much"; one must also ask what the combustion is being exchanged for**—the unit of electricity saved by using AI to optimize the grid and the unit of electricity combusted to train some large model are the same unit of electricity in thermodynamics—the physical quantities remain comparable—though they support different kinds of uses. The former is computation placed where it can stand, doing what it can do; the latter, if it pushes computation in the direction of "letting machines impersonate judgment, impersonate understanding, impersonate care," falls under what Weizenbaum's criterion calls the first kind—the substitution of human functions involving interpersonal respect, understanding, and love—and existing aggregate units (kilowatt-hour, tonne of carbon, dollar of GDP) cannot, ex ante, distinguish the use value of these two types of electricity. This is not a matter of insufficient measurement precision—output heterogeneity limits the decision-making sufficiency of a single metric, not the arithmetic of comparable physical quantities; it is precisely what §7's "impossibility of accounting" seeks to characterize, first appearing in more concrete form in §6's "fracture point"—why the aggregation tools that enabled the tobacco model to function are no longer sufficient for AI: linear extrapolation from per-use costs alone cannot infer system consequences; synchronized loads, investment inertia, and demand feedback can invalidate that simplification, requiring governance at the infrastructure and investment levels without making aggregation or consumer-facing tools inherently impossible.

Thus this chapter is not an indictment of AI; it is identifying a more precise matter: all deployments beyond that boundary jointly marked by Weizenbaum and Dijkstra—keeping the computer in the position where it can stand—are today being quietly advanced by a ledger that no existing account adequately represents, and the bill is being paid with the atmosphere, freshwater, and the biosphere. The grammatical subject of the decision is locked back to the deployer: which institutions, which capitals, which discourses are choosing to push computation into positions beyond that boundary; whose decisions are causing this combustion to occur, and whose rhetoric is granting this combustion legitimacy.

§6 The Fracture Point

It is now possible to state precisely where the tobacco analogy fractures.

Here, an existing academic framework must first be situated, lest this chapter's "fracture" argument be misread as a denial of the tobacco–AI continuity. The concept of **limbic capitalism** proposed by Courtwright

(2019) places tobacco, alcohol, sugar, high-sugar high-fat processed foods, gambling machines, social media, smartphones, and the various "habit-forming" applications within them on a single historical-structural spectrum, diagnosing their shared operational logic—"a technologically advanced but socially regressive business system" whose products "create (initially) intense brain reward and foster destructive habits in consumers"; he further points out that the Internet is "their most radical technological expression" of this system (Courtwright, 2019). Within Courtwright's genealogy, from the mechanized cigarette of the nineteenth century to the habit-forming product design of the twenty-first century à la Nir Eyal (Habit Summit, "customer lifetime value," "short viral-cycle time"), there is a continuous techno-commercial evolutionary line, not isolated phenomena separated from one another.

This chapter makes its argument within this genealogy and does not deny this continuity: the AI infrastructure expansion depicted in the preceding §§2–5 continues forward along the same trajectory of limbic capitalism as defined by Courtwright—it inherits the entire commercial logic of creating rewards in the brain to lock in use and couples it with a physical infrastructure of unprecedented scale.

However, **the fracture point this chapter seeks to identify lies not at the level of "whether it still belongs to the same genealogy"—it does—but at the level of the applicability of governance tools**. The policy imagination Courtwright himself relies upon in *The Age of Addiction* remains essentially the traditional public-health–anti-vice framework: taxation, advertising restrictions, protection of minors, addiction treatment, regulation of product formulation and design (Courtwright, 2019). These tools can reduce harm by influencing adoption, consumption intensity, and product design. The issue is not that they require weakly correlated loads, but that per-use costs and their linear extrapolation may inadequately represent infrastructure investment, lock-in, and demand feedback. AI therefore requires governance at the infrastructure and investment levels as well; its externality structure is no longer of the aggregation type, but of the emergence type. Therefore, when this chapter says "the tobacco analogy fractures here," what fractures is not the qualitative distinction of "addictive–non-addictive" (Courtwright has amply demonstrated that this distinction itself is continuous), but the **shrinkage of the applicability of aggregation tools after crossing a certain threshold of scale and coupling—linear extrapolation is no longer sufficient, and the scale of governance must expand**.

What follows unfolds this categorical shift.

Tobacco's externality structure is an aggregation model: a large number of individuals make harmful choices → the sum of individual health damages → public health burden. The behavior of the system is the sum of the behavior of its components—the public health cost of one hundred million smokers is approximately one hundred million times the cost of one smoker. In this model, the intervention strategy is clear: alter individual behavior (through taxation, bans, information disclosure) → reduce the total number of harmful choices → linearly reduce the public burden. This intervention strategy is effective because the level at which the externality is generated (individual consumption behavior) and the level at which the intervention acts (individual consumption behavior) are the same level.

AI's externality structure is an emergence model: investment decisions at the infrastructure level → self-reinforcing expansion dynamics → systemic pressure on planetary boundaries. The behavior of the system cannot be reduced to the sum of individual behaviors, because the four reinforcing loops and the Jevons mechanism create non-additive couplings between individual behaviors (more precisely, between individual behaviors and infrastructure decisions). In this model, interventions at the individual level—making users

"use AI less," labeling AI queries with "carbon footprints," encouraging "green AI usage habits"—are not wrong, but on their own they may fail to constrain committed investment and multiyear infrastructure lock-in in time. One can make one million users each use 10% less AI, but this may not immediately alter committed construction plans, and may affect subsequent investment only through revenue expectations—because the construction decision is not driven solely by the current marginal demand of individual consumers, but by R4 (Economic Lock-in) and R5 (Infrastructure Lock-in), two loops that operate downstream of individual consumption behavior and possess their own independent inertia; infrastructure lock-in delays the transmission of demand changes without eliminating it. Reducing individual usage in the face of AI infrastructure expansion is like using a cup to bail water in the face of a flood. Not that one shouldn't bail, but the rate of bailing and the rate of inflow are not on the same order of magnitude.

This paradigm shift has a precise isomorphic formulation in the academic discussion within power systems engineering. Bashir et al. (2026) directly diagnose the impact of AI training workloads on the grid as a violation of the **load diversity** assumption—an assumption that has anchored grid planning for nearly a century, its core being that "the uncorrelated demands of millions of small consumers produce a smooth, predictable aggregate"—correlations between loads are low enough that aggregate fluctuation is far smaller than the sum of individual fluctuations; grid planning relies on this aggregate smoothing, not on strict independence between loads. AI training campuses weaken the smoothing condition provided by low synchronicity: the parallel jobs of thousands to tens of thousands of GPUs within a single campus are coordinated in start-stop and checkpointing by a single scheduler, presenting themselves at the power level as a single, concentrated load spanning hundreds of megawatts to the gigawatt level. Without adequate facility-level buffering and workload coordination, these synchronized power swings can propagate to the grid connection; their magnitude and timescale depend on power-supply, storage, and control arrangements (Bashir et al., 2026). The entire toolkit on which grid planning has relied for nearly a century—from the diversity factor to daily peak load forecasting to capacity planning—is built upon the assumption of aggregate smoothing; when loads within a campus are highly synchronized, planners cannot assume without validation that other loads will sufficiently smooth its power fluctuations—smoothing across campuses remains dependent on inter-campus correlations, common disturbances, and scheduling arrangements. This diagnosis is structurally isomorphic with the failure of the tobacco–AI analogy in this chapter: the low-correlation premise on which grid aggregate smoothing depends is dissolved by the **coupling structure** within the system—but this diagnosis targets grid-planning tools and does not automatically extend to governance tools aimed at individual behavior or emissions sources, such as taxes and disclosure.

The fracture point is here: the tobacco analogy understands AI's externality as a "summation problem"—one need only reduce the total volume of harmful individual behaviors—whereas AI's externality is in fact a "structure problem"—the loop structure of the system itself needs to be changed. When the tobacco framework is applied to AI's environmental impact, attention is directed toward individual carbon footprints, the energy consumption of a single query, the "responsible use" of consumers—these are all projections of linear thinking onto a nonlinear system. Such interventions give the feeling of doing something, but the level at which they intervene and the level at which the problem is generated are not on the same plane.

The reason this fracture point is "the heart of the matter" is not only that the tobacco framework fails here—every framework has boundaries of applicability—but that if one remains within the tobacco framework without self-awareness, the scale of the problem will be systematically underestimated, and limited policy attention will be

invested in intervention schemes structurally insufficient to address the problem. In the simplified consumption-side model adopted in this chapter, what the tobacco framework yields is a solution to a public health problem. What this chapter confronts is an Earth system problem. Between the two lies not a difference of degree, but a categorical shift. Public health problems have clear causal chains, quantifiable harms, attributable responsible subjects. Earth system problems have feedback loops, tipping points, irreversible phase changes, and attributability that vanishes within the causal chain. When AI's physical footprint crosses the threshold of this categorical shift—when it transforms from "the carbon emissions of one industry" into one of the "systemic drivers of planetary boundaries"—what is needed is not better tobacco regulation, but an analytically and governmentally different paradigm.

Beyond infrastructure-level coupling, the software-engineering literature identifies another accounting-relevant piece of mechanism evidence: functioning and deliverable output may conceal reliability problems and long-term maintenance costs. Ward Cunningham, in 1992, while developing a cash-securities portfolio management system (WyCash+) for U.S. pensions, corporations, and banks, first wrote into academic language the phenomenon of unrefactored code accumulating as debt, and pointed out that "immature code may work fine and be completely acceptable to the customer . . . Entire engineering organizations can be brought to a standstill under the debt load of an unconsolidated implementation" (Cunningham, 1992). The provenance of this diagnosis means: Cunningham's debt analogy, the Sculley-Ernst engineering corroboration, and the financial diagnosis of Chapter 22 are structurally isomorphic in the delayed exposure of costs—structural isomorphism is not identity of mechanism.

Twenty-three years later, Sculley et al. (2015), at the Conference on Neural Information Processing Systems (NeurIPS), extended this framework, pointing out that machine learning systems bear a kind of **hidden** technical debt: not debt at the code level, but system-level entanglement, hidden feedback loops, undeclared consumers, and data dependencies. Ernst, Kazman, and Delange (2021, §9.1) further made precise the **default failure mode** of ML systems: unlike traditional software that crashes or halts, "machine learning systems will usually continue to function, but do so inadequately"—execution has not stopped, but the result is wrong, and degraded accuracy is harder to detect than obvious crashes. This is precisely the independent engineering testimony of the **pseudo-small-fiber** defined in Chapter 4 §2.4: the true fiber is large (the system is wrong) while the perceived fiber is small (the interface continues to output; the customer continues to pay); the difference between the two is the Earth being combusted. And all these observations already had an early case in the 1966 Weizenbaum secretary incident—his secretary "surely knew it to be merely a computer program . . . After only a few interchanges with it, she asked me to leave the room" (Weizenbaum, 1976)—the anthropomorphic projection onto pseudo-small-fibers is not a new pathology of the large language model (LLM) era; it is a sixty-year-old disease.

Synthesizing these four layers of evidence, this chapter's fracture point can therefore be given its precise name: **under the current loop structure, what the Earth combusted by AI infrastructure is exchanged for includes pseudo-small-fibers and sycophancy—unmaintainable immature code that looks like production, the substitution of a sense of completion for completion, Rogerian reflection that looks like therapy, mechanical computation that looks like judgment; their share of total output is unmeasured, yet the materials above independently attest, layer by layer, to the same mechanism—continued output does not establish output reliability, and errors and maintenance costs can stay hidden**—it is one of the intermediate terms through which the loop structures accumulated in §§3–5 lead to the accounting collapse of §7.

And at present, the fact that this categorical shift has already occurred has not even been clearly recognized.

§7 The Impossible Accounting

The accounting collapse diagnosed in §6 of this chapter is not an isolated event; it has a thirty-year engineering prehistory within the financial system. "Impossible accounting" here does not mean that electricity use, water consumption, and emissions cannot be measured. It means that a single accounting metric cannot, by itself, establish all environmental consequences or justify deployment. The sections below distinguish allocation rules, dynamic feedback, and tipping risks, rather than treating them as a failure of physical measurement. The scene in which Cunningham (1992) proposed "technical debt" was itself a piece of bookkeeping software serving a multi-billion-dollar cash-securities portfolio management system—the four words debt, interest, repayment, and grinding to a halt simultaneously bear engineering semantics and financial semantics in his text; Cunningham himself called this mechanism a "debt analogy," but the structure of delayed cost exposure that the analogy points to is precisely what this chapter tracks. Chapter 22 has already diagnosed the three governance-clock modes of the financial system after shadow debt $D_c^{shadow} = D_c^{true} - D_c^{obs}$ accumulates to the critical point (the long debt clock of 1990s Japan, the crisis-driven critical phase transition of the U.S. in 2008, the incremental manifestation of China's 2018 Asset Management Regulations), the core mechanism being the phase transition of the manifestation function $M(\tau)$ at the critical point—debt itself is not the problem, **the prolonged uncalibrated gap between the debt and the ledger** is. The accounting collapse of AI infrastructure (the triple collapse to be unfolded below) is structurally isomorphic to the financial calibration mechanism in the delayed exposure of costs—but this time it is not the same ledger: the financial system kept its accounts in monetary terms, while the electricity use, water consumption, and emissions of AI infrastructure are already recorded in part; the problem is that these records may not adequately inform deployment decisions, nor capture delayed damage, distributional consequences, and output reliability. What follows diagnoses this triple collapse one by one.

This chapter has no solutions. Not out of humility, but because the problem structure described in this chapter implies that any solution operating entirely within existing institutional frameworks is most likely insufficient.

But it is possible to point out the precise manner in which existing frameworks fail—because the diagnosis of failure modes is itself useful.

The standard tools currently used to address environmental externalities are pricing instruments—carbon taxes and emissions trading—alongside procedural tools such as environmental impact assessment. Carbon taxes set an emissions price, while cap-and-trade sets an emissions budget and allows trading to establish prices; both involve accounting boundaries and policy objectives, but neither necessarily requires a complete monetization of every environmental loss. Environmental impact assessment is a procedural review and does not presuppose monetization. This requirement of accounting boundaries holds roughly in linear systems: the social cost of one tonne of carbon emissions can be estimated (even if the range of estimates spans an order of magnitude); the emissions of a factory can be attributed to its products; consumers adjust behavior via price signals.

In the loop structure of AI, this accounting collapses on three levels.

Attributional Collapse. What is the carbon footprint of a single AI query? This question appears to have a definite answer, but it is in fact an allocation problem, and the choice of allocation rule is not uniquely determined by physical facts. How are training costs amortized? A model is trained once and then deployed to answer billions

of queries—the method of amortizing training costs (by time? by number of queries? by economic value?) will drastically alter the "carbon footprint" of a single query. How is the embedded carbon of infrastructure handled? Should the carbon emissions from the steel, cement, and chip fabrication used to build the data center be included in the carbon footprint of an AI query? If so, in what proportion? How is the marginal grid carbon intensity attributed? Is the increased electricity demand due to AI met, at the margin, by newly added renewable energy or by fossil fuel peaker plants? The answer depends on region, time of day, grid dispatch logic, and the degree of excess capacity—physical facts alone cannot determine all allocation rules. How are indirect effects handled? AI improves the efficiency of some industry, leading that industry to expand, and the expansion generates new emissions—are these emissions attributed to AI? If so, how much? At every fork in the causal chain, an allocation decision must be made that is not fully determined by physical facts—allocation rules are not unique, but each rule must state its boundaries and sensitivity; a uniquely determined "physically correct" attribution does not exist, while reviewable attribution methods do, and the carbon footprint of the same AI query may vary substantially with the allocation rule and boundary choices. Precise attribution is not in principle impossible, yet existing product carbon-footprint accounting alone does not adequately reflect these systemic consequences—complexity that is not accidental but an intrinsic feature of a coupled system. A distinction of accounting levels is also needed: allocation choices complicate per-query lifecycle footprints, but do not necessarily prevent regulation at the fuel or emissions-source level—the limits of attributional collapse chiefly affect product comparisons, responsibility claims, and deployment judgments, and cannot be equated with emissions regulation itself being inoperable.

The deeper problem of attributional collapse lies not only in technical complexity but also in the existence of institutionalized pathways for exemption tools with parallel precedents in the history of climate governance. Crawford (2022) points out that even within the mature field of climate accounting—which has operated for thirty years since its inception and has accumulated a complete conceptual system of Scope 1/2/3 and IPCC methodologies—military greenhouse gas emissions are still disclosed primarily in forms far narrower in scope than actual emissions: the U.S. military's own overseas operational emissions, emissions generated in adversary territory as a result of U.S. military operations abroad, and Scope 3 emissions from the military-industrial supply chain are mostly either undisclosed or handled separately in ways incompatible with mainstream climate accounting methodologies (Crawford, 2022). In other words, even within the most mature, most internationally consensual domain of carbon accounting, **a single institutional gap has sufficed to leave the carbon emissions of the largest single institution on the planet without comprehensive, transparent accountability**. And the attributional collapse of AI infrastructure described above far exceeds the military emission accounting of 1997 in terms of coupling complexity, while its intimate stitching with the national security narrative (unfolded in §3 B2) overlaps precisely with the institutional template diagnosed above. An institutional template historically verified as effective, plus a new category of emission source technically even harder to track, represents, in the most pessimistic scenario: attributional collapse need not await technical failure; it already has an institutional exit path prepared.

Temporal Collapse. Carbon pricing typically discounts and sums damages across periods—but adding across periods does not itself require damages to be linear in emissions: with lock-in and feedback, today's behavior alters the trajectory of future emissions itself. But in a system with positive feedback loops and infrastructure lock-in effects, a data center built today does not only produce its own construction and operational carbon emissions. It raises, through the R5 loop (Infrastructure Lock-in), an inertial baseline of emissions for the

next 15 to 25 years—one cannot "un-build" a data center whose concrete has already been poured, and exit costs make continued operation the baseline scenario. It generates, through the R1 loop (Capability–Demand Spiral), demand for more data centers—each new facility provides the justification for the next. Pricing such path-dependency effects requires not a simple carbon price, but a dynamic framework capable of modeling loop structures and lock-in effects—this far exceeds the informational capacity that a carbon tax can carry. A carbon tax, if used only statically—at too low a level, under a non-credible future price path, or with relevant costs left outside the decision—will struggle to alter long-term investment; path dependence demands scrutiny of price levels, persistence, and coverage, not the abolition of the price instrument itself. Evaluating infrastructure lock-in requires more than construction-period emissions and prices: it must consider future operating emissions, expected price paths, exit costs, and investment responses. The issue is whether actual institutions cover these factors adequately and credibly, not whether intertemporal pricing necessarily assumes linear damages or must charge all future emissions again through the price of a current tonne.

Boundary Collapse. The theoretical basis of carbon pricing is marginal analysis—at the current emission level, what is the social cost of adding one more tonne of emissions? Conventional derivative-based marginal analysis using smooth damage functions depends on differentiability assumptions. But in the processes characterized by the planetary boundaries framework, tipping transitions may introduce sharp nonlinearities and irreversible losses, with the associated damage function depending on the processes modeled and the treatment of uncertainty. Near a tipping point, the social cost of the marginal tonne of carbon may rise sharply, with deep uncertainty in both probability and consequence; societies may also impose non-negotiable limits on certain irreversible losses. These reasons support constraints beyond pricing—aggregate caps, irreversibility prohibitions—but they do not establish that marginal costs tend to infinity, nor that all economic analysis fails: shifts such as the collapse of the Greenland ice sheet, the shutdown of the Atlantic Meridional Overturning Circulation, or large-scale dieback of the Amazon rainforest have been incorporated into some existing economic models, but their valuations do not exhaust the social and ethical significance of irreversible losses. This is precisely the boundary of price instruments: price is a tool for estimating known damages and cannot bear the full valuation of tipping losses whose probability, magnitude, and recoverability are all deeply uncertain.

The fiber framework provides a unified diagnosis of this triple collapse. In this chapter's fiber framework, the problem is neither that emissions cannot be measured nor that all pricing instruments require tradable allowances. It is the treatment of a bounded carbon metric as a sufficient representation of a deployment's full consequences and legitimacy—the information loss arises primarily from reducing multidimensional consequences to a single metric, not from quantification or discretization itself. But AI's material footprint is not a small-fiber clock. It is a measurement object of extremely large fiber, and its fiber size grows as the mapping chain lengthens (whether superlinearly must be assessed for specific mappings and metrics). Attributional collapse is a direct consequence of fiber inflation—the mapping chain (from ore, energy, water to GPU, data center, AI output) is too long, too complex, and the associations between nodes are dynamic and nonlinear. Temporal collapse is a case of dynamic corrosion—the behavior being priced (today's emissions) alters, through technological lock-in, future behavior (subsequent emissions carry inertial continuation). Boundary collapse concerns the reduction of multidimensional consequences—Earth-system outcomes involving thresholds, uncertainty, and distributional effects are compressed into a carbon or price metric that is insufficient on its own to justify deployment. The three are not three parallel difficulties, but three surfaces of the same structural problem: the difficulty facing carbon pricing is not that carbon emissions are hard to measure, but that attribution, feedback, and tipping

risk mean that the correspondence between "measurable physical quantities" and "accounts that can ground decisions" is no longer guaranteed—a carbon metric defined within an explicit accounting boundary can support regulation, but does not by itself establish that other material costs, distributional consequences, and the value of the deployment have been adequately addressed.

This triple collapse—attributional, temporal, and boundary—means that carbon pricing, the most widely accepted governance tool for environmental externalities, is not only practically difficult to operate but also theoretically insufficient when confronting the loop structure of AI and the nonlinear dynamics of planetary boundaries. This is not to say that carbon pricing should be abandoned—within its scope of applicability, it remains a useful tool. But its scope of applicability does not cover the core of the problem described in this chapter.

If pricing tools are insufficient, what would suffice? This chapter does not propose a complete solution. But it can point out the constraints any sufficient governance framework must satisfy: any adequate governance framework must be able to intervene in the loop structure itself (rather than only in the outputs of the loop), handle nonlinearity and tipping points (rather than only marginal changes), and span attributional ambiguity (rather than being paralyzed by it). This rules out most existing tools. The directions it may point toward include: direct scale regulation of AI infrastructure expansion (the logic of a quantity constraint—the side that fishery quotas and cap-and-trade share—rather than a pure price signal), imposing rate limits on the key loop nodes of AI R&D (like circuit-breaker mechanisms in financial regulation), or—most radical and most unimaginable—imposing a global cap on total AI computation and allocating quotas among nations, analogous to the logic of a carbon budget. Any one of these directions is nearly unrealizable in the current political-economic landscape—but "unrealizable" does not equal "unnecessary." If the structure of the problem demands a solution of a certain scale, and a solution of that scale is currently unimaginable, then what needs to change is not the diagnosis of the problem, but the boundaries of the imagination.

§8 Combustion

Return to the first sentence of this chapter.

This sentence is burning. The paragraph you are reading now is also burning. Every interaction with any AI today is burning. "Burning" is this chapter's shorthand for material consumption and dissipation; where fossil energy supplies the computation, it also refers to literal fuel oxidation—and that reaction is currently providing, in a manner imperceptible to humans, in places invisible to humans, a portion of the energy for this reading experience.

Chapter 28 of this book discussed how AI, through sycophancy, distorts human cognition of the world. Chapter 31 discussed how Agentic AI converts readings directly into physical execution—the transmission structure's full impact on the human defense side (Chapter 31's relational-morphology analysis belongs to the cultural layer of the book's threefold division; its transmission structure is the executive tip of that layer reaching into the material). What this chapter discusses is another obscured dimension: how AI operations impose changes on the physical world whose consequences are hard to reverse. These three levels—the cognitive, the cultural, the material—are not three parallel problems, but three cross-sections of the same system, and positive feedback exists among them. AI's sycophancy makes people use it more frequently (cognitive lock-in → more inference → more burning); AI's companionate design makes it harder for people to detach from it (relational dependency → longer conversations → more burning); and more burning—more users, more inference, more

revenue—provides AI companies with larger-scale capital, driving larger-scale infrastructure investment, raising the inertial baseline of future energy consumption. The positive feedback loops at the cognitive, cultural, and material levels are not operating independently; they are mutually nested and mutually reinforcing. Algorithmic Xiangyuan is not only an epistemological problem—it is also a thermodynamic problem: every superfluous act of ingratiation has a carbon footprint.

Beyond these three loops (cognitive, cultural, material), a fourth must also be identified: **the automation paradox loop**. This mechanism has its classic prototype in the aviation industry—pilots' long-term dependence on autopilot leads to the degradation of manual flying skills, thus further locking them into autopilot. Manual takeover is most needed precisely when automation has failed. The 2009 Air France AF447 crash is the most famous disaster sample of this loop—unreliable airspeed indications associated with ice-crystal obstruction of the pitot probes were followed by autopilot disengagement and a transition to manual control; the aircraft subsequently entered a stall from which the crew did not recover; 228 people perished. Not until the Paris Court of Appeal's ruling of May 21, 2026 was the chain of responsibility pulled back, at the criminal level, onto the manufacturer and the airline—prosecutors' allegations included the failure to follow up on earlier sensor flaws and poor training—rather than being compressed simply into the pilots' errors in the last few minutes (Reuters, 2026).

The same structure repeats itself in professions where AI substitutes for human output—Chapter 35 §2.3.5 has already diagnosed the contemporary form of this process as "the intensification of cognitive substitution"—platforms extending the centralized processing of information distribution to the centralized processing of cognitive production itself: AI replaces humans → the corresponding human skills atrophy → humans become more dependent on AI → the scale of AI deployment expands → energy consumption and physical footprint rise again. The peculiarity of this loop is that the eroded stock of judgment cannot be reproduced by increasing the electricity supply (Chapter 35 §8.3)—the recoverability of skills depends on training and organizational conditions, but rebuilding the stock takes time, and the combustion has already occurred. If the cognitive–cultural–material triple loop still leaves room for balancing loops (reduce usage, change design, switch energy sources), the fourth loop erodes, at the source, the very premise that "one can use AI less"—when the stock of humans capable of independent judgment and operation outside of AI is continuously diluted, "using less" ceases to be an option.

The tobacco analogy was once a powerful heuristic framework. But frameworks have boundaries of applicability. What tobacco burns is a cigarette; what it damages is a pair of lungs. AI's material substrate spans the oxidation of fossil carbon, the fission of uranium, and the manufacture of silicon—three physical processes, one externalized cost chain—and what it affects is the planetary atmosphere, the water cycle, and material flows. Between these two costs lies not a difference of quantity, but a structural transformation—from linear externalities to coupled externalities, from individual damage to systemic overshoot risk, from priceable costs to risks not adequately represented in existing accounts.

Recognizing this will not stop the combustion. But it at least eliminates a dangerous illusion: this is not a problem solvable by putting carbon labels on AI queries, just as climate change is not a problem solvable by having everyone drive one day less. The problem generation occurs at the system level—in the coupling of reinforcing loops, in the lock-in of infrastructure, in expansionary mechanisms that may keep strengthening—and the solution, if there is one, must also be at the system level.

And at present, a framework capable of even posing the right questions at the system level has not yet been fully established. One cannot find on a map a place the map has not yet drawn, and the current mainstream discourse on AI's environmental impact—whether the "sustainability" reports of technology companies or the initiatives of environmental organizations—is still using a map drawn for linear externalities to navigate a nonlinear terrain.

What this chapter attempts to contribute is merely some contour lines of that new map. In a system of exponential—let alone super-exponential—growth, the lateness of a framework and the absence of a framework may not differ much in consequence.

The book Weizenbaum wrote at MIT in 1976 closes with a monologue: what is truly eroded by the temptation of computation is not the Earth, but the human—**it enslaves the mind that has no other metaphors and few other resources to call on** (Weizenbaum, 1976). The imperialism of instrumental reason he named is not equivalent to opposing reason—he explicitly stated, "I combat the imperialism of instrumental reason, not reason." The entire argument of this chapter merely lands this naming onto the material layer: the first thing combusted is not coal, not water, not the atmosphere outside the cooling tower—it is that **before the combustion occurred, the human had already self-mechanized**. The human first degraded their own judgment into calculation, their own care into information processing, their own labor into a replaceable function, and only then permitted a physical infrastructure to be built to execute that degradation. The criterion Weizenbaum himself gave in 1976 was already sharp enough: the first kind—applications substituting for human functions involving interpersonal respect, understanding, and love ("obscene")—should not be undertaken; the second kind—applications with irreversible and not entirely foreseeable side effects—should be avoided, or at least not undertaken without very careful forethought; and ought not to be pursued if, in addition, they cannot be shown to meet a pressing human need that cannot readily be met in any other way (Weizenbaum, 1976). Fifty years later, an 835-megawatt unit slated for restart, a parallel cluster of tens of thousands of GPUs, an AI training campus waiting years for grid interconnection—these fall under the first half of the second kind: their side effects are irreversible and not entirely foreseeable. Whether they also fall under the first kind depends on the specific uses those facilities support—precisely the question that facility scale should not answer; and the second half of the second kind—whether this combustion is exchanged for a pressing need that cannot readily be met otherwise—this chapter has explicitly left suspended in §5.

And at this moment, it is still burning.

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